

Siddhartha Dhiman

SENIOR SOFTWARE ENGINEER · FLYWHEEL

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🐼 TheJaeger | 🌐 [siddharthadhiman](#) | 🎓 Google Scholar | 🇸🇬 Singapore Citizen · 🇺🇸 US Permanent Resident



Senior Software Engineer | Python | Distributed Systems | Cloud-Native Applications | Medical Imaging Platforms

Summary

Senior Software Engineer with 8+ years of experience building cloud-native backend systems for healthcare and biomedical research platforms. Specialized in distributed systems, microservices architecture, and large-scale data movement. Proven track record delivering reliable Python services that process and transfer hundreds of terabytes of medical imaging data across AWS, Azure, and Google Cloud environments. Experienced in system design, technical leadership, and mentoring engineers while delivering scalable, production-grade software.

Authorized to work in the United States and Singapore.

Skills

Languages	Python, SQL, Bash
Databases & Storage	PostgreSQL, MongoDB, Redis
Backend Engineering	FastAPI, REST APIs, Distributed Systems, Microservices Architecture
Cloud & Containers	Docker, Kubernetes, Helm, AWS, Azure, Google Cloud Platform (GCP)
Engineering Practices	System Design, CI/CD, Observability, Testing, Performance Optimization
Domain Expertise	Medical Imaging, DICOM, Neuroimaging, Biomedical Data Platforms, Clinical Trials

Experience

Flywheel.io

Remote

SENIOR SOFTWARE ENGINEER

Jan. 2023 - PRESENT

- Own the design, development, and operational reliability of backend services powering Flywheel's biomedical imaging platform, supporting large-scale data ingestion, transfer, and management workflows.
- Lead development of cloud-native Python microservices deployed to Kubernetes, enabling scalable and resilient processing of medical imaging datasets across distributed systems.
- Design and implement high-throughput data transfer services responsible for securely moving hundreds of terabytes of imaging data between imaging modalities, healthcare systems, cloud storage providers, and customer environments, cloud storage providers, and customer environments.
- Drive technical decisions across service architecture, API design, data modeling, observability, performance optimization, and operational excellence.
- Architect and maintain reusable Helm-based deployment frameworks that streamline application delivery and improve deployment consistency across Kubernetes environments.
- Integrate backend services with AWS, Azure, Google Cloud Platform, and healthcare imaging systems to support diverse customer workflows and data management requirements.
- Optimize data-intensive applications through strategic use of PostgreSQL, MongoDB, and Redis, improving performance, reliability, and scalability.
- Collaborate closely with product, infrastructure, and customer-facing teams to translate complex business requirements into scalable technical solutions.
- Mentor engineers through code reviews, architectural discussions, and technical guidance while promoting engineering best practices across the organization.

Medical University of South Carolina (MUSC)

Charleston, SC

SYSTEMS PROGRAMMER/DEVELOPER

Sep. 2018 - Dec. 2022

- Led development of scalable scientific-computing pipelines responsible for processing, analyzing, and transferring over 100 TB of neuroimaging data across distributed computing environments.
- Co-created PyDesigner, an open-source diffusion MRI processing framework adopted by research institutions worldwide, improving reproducibility and accessibility of advanced neuroimaging workflows.
- Designed and implemented machine learning and deep learning solutions for neuroimaging applications, including epilepsy prediction, tissue classification, and automated image segmentation.
- Built reproducible, containerized research environments using Docker and Singularity, reducing deployment friction and improving portability across heterogeneous infrastructure.
- Co-developed novel white matter imaging methodologies, including Fiber Ball Imaging (FBI) and Fiber Ball White Matter Modeling (FBWM), contributing to advances in brain microstructure analysis.
- Collaborated with clinicians, neuroscientists, and researchers to translate scientific objectives into scalable software solutions and actionable insights.
- Mentored students and junior researchers on software development, scientific computing, and data analysis best practices.
- Authored peer-reviewed publications and maintained production-grade software supporting advanced neuroimaging research initiatives.

State University of New York at Buffalo

Buffalo, NY

DATA ANALYST (INCL. MASTER'S THESIS)

Jan. 2017 - May 2018

- Performed statistical analysis of RNA sequencing datasets using correlation analysis, ANOVA, and Kolmogorov-Smirnov testing to identify disease-associated gene expression patterns.
- Developed image-processing and segmentation workflows for biomedical research using feature engineering, clustering, and computer vision techniques.
- Evaluated emerging machine learning approaches, including convolutional neural networks, for large-scale biomedical imaging applications.
- Implemented image-analysis algorithms including watershed segmentation and supervoxel-based clustering to characterize vascular and cellular structures in neurological and histopathological datasets.
- Applied machine learning and quantitative analysis techniques to support Alzheimer's disease, schizophrenia, and diabetic nephropathy research initiatives.

Software

PyDesigner

MUSC

COMPLETE DTI/DKI/WMTI/FBI/FBWM PRE-PROCESSING PIPELINE WRITTEN IN PYTHON

2018 - PRESENT

- Co-created the platform and developed scalable image-processing workflows to improve reproducibility and accessibility of advanced neuroimaging analysis.

dTractSeg

MUSC

TRACKSEG COMPATIBILITY LIBRARY THAT ENABLES SEAMLESS INTEGRATION BETWEEN PYDESIGNER AND TRACTSEG

2018 - PRESENT

- Completely converts PyDesigner outputs and executes TractSeg in one line
- Enables easy AFQ-styled study analysis on PyDesigner processed data

Publications

Systems Genome: Coordinated Gene Activity Networks, Recurring Coordination Modules, and Genome Homeostasis in Developing Neurons

State University of New York

INTERNATIONAL JOURNAL OF MOLECULAR SCIENCES

May. 2024

- **Siddhartha Dhiman**, Namya Manoj, Michal Liput, Amit Sangwan, Justin Diehl, Anna Balcerak, Sneha Sudhakar, Justyna Augustyniak, Josep M. Jornet, Yongho Bae, Ewa K. Stachowiak, Anirban Dutta, Michal K. Stachowiak

Comparison of two types of microscopic diffusion anisotropy in mouse brain

MUSC

ANNALS OF NEUROLOGY

Aug. 2022

- Jens H. Jensen, Josh Voltin, Xingju Nie, **Siddhartha Dhiman**, Emile T. McKinnon, Maria F. Falangola

Greater Diffusion Restriction in White Matter in Preclinical Alzheimer Disease

MUSC

ANNALS OF NEUROLOGY

Mar. 2022

- Andreana Benitez, Jens H. Jensen, Kathryn Thorn, **Siddhartha Dhiman**, Stephanie Fountain-Zaragoza, William J. Rieter, Maria Vittoria Spampinato, Eric D. Hamlett, Paul J. Nietert, Maria de Fatima Falangola, Joseph A. Helpert

Fiber Ball white matter modeling reveals microstructural alterations in healthy brain aging

MUSC

AGING BRAIN

Feb. 2021

- **Siddhartha Dhiman**, Stephanie Fountain-Zaragoza, Jens H. Jensen, Maria Fatima Falangola, Emilie T. McKinnon, Hunter G. Moss, Kathryn E. Thorn, William J. Rieter, Maria Vittoria Spampinato, Paul J. Nietert, Joseph A. Helpert, Andreana Benitez

PyDesigner: A Pythonic Implementation of the DESIGNER Pipeline for Diffusion Tensor and Diffusional Kurtosis Imaging

MUSC

BIORxiv

Oct. 2021

- **Siddhartha Dhiman**, Joshua B Teves, Kathryn E Thorn, Emilie T McKinnon, Hunter G Moss, Vitria Adisetiyo, Benjamin Ades-Aron, Jelle Veraart, Jenny Chen, Els Fieremans, Andreana Benitez, Joseph A Helpert, Jens H Jensen

Diffusion MRI detects early brain microstructure abnormalities in 2-month-old 3x Tg-AD mice

MUSC

NMR IN BIOMEDICINE

Jun. 2020

- Maria Fatima Falangola, Xingju Nie, Ralph Ward, Emilie T. McKinnon, **Siddhartha Dhiman**, Paul J. Nietert, Joseph A. Helpert, Jens H. Jensen

Computational analysis of cerebrovascular structures imaged using two-photon microscopy

State University of New York

SPIE MEDICAL IMAGING: DIGITAL PATHOLOGY

Mar. 2019

- **Siddhartha Dhiman**, Itender Singh, Pinaki Sarder

Role of nuclear-FGFR1 in schizophrenia and ontogenic genome programming – a computational study

State University of New York

PROQUEST

Jun. 2018

- **S. Dhiman**

Cerebral organoids reveal early cortical maldevelopment in schizophrenia—computational anatomy and genomics, role of FGFR1

State University of New York

NATURE: TRANSLATIONAL PSYCHIATRY

Nov. 2017

- E. K. Stachowiak, C. A. Benson, S. T. Narla, A. Dimitri, L. E. bayona Chuye, **S. Dhiman**, K. Hari Krishnan, S. Elahi, D. Freedman, K.J. Brennan, P. Sarder, M.K. Stachowiak

Program Committees

2013 **Secretary**, Taiwanese Student Association

State University of
New York

Education

State University of New York at Buffalo

Buffalo, NY

M.S. IN BIOMEDICAL ENGINEERING

Aug. 2016 - June. 2018

- CGPA: 3.804/4.000

State University of New York at Buffalo

Buffalo, NY

B.S. IN BIOMEDICAL ENGINEERING

Aug. 2012 - May. 2016

- Awarded the International Admissions Scholarship